Used Cars Price Analysis

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1. INTRODUCTION

The used car market has grown significantly in recent years due to increasing vehicle prices, rapid technological advancements, and changing consumer preferences. Many buyers prefer pre-owned cars because they are cost-effective and provide better value for money compared to new vehicles.

This project focuses on analyzing used car prices by collecting data from reputed online platforms through web scraping. The analysis aims to identify pricing trends, understand the impact of factors such as brand, fuel type, manufacturing year, and distance driven, and help buyers make informed purchasing decisions.

By applying data cleaning, exploratory data analysis (EDA), and visualization techniques, this project transforms raw data into meaningful insights that highlight market behavior and pricing patterns.

2.PROBLEM STATEMENT

The price of used cars varies widely depending on several factors such as brand reputation, vehicle age, fuel type, and usage. Buyers often struggle to determine whether a car is fairly priced.

This project addresses the following problems:

- Understanding how car prices vary across different brands.
- Identifying high-quality cars available at lower prices.
- Analyzing the influence of fuel type, distance driven, and manufacturing year on pricing.
- Providing data-driven insights to support smarter purchasing decisions.

3.OBJECTIVES

The primary objectives of this project are:

- To collect used car data using web scraping from trusted platforms.
- To clean and preprocess the dataset for accurate analysis.
- To perform exploratory data analysis to uncover trends and patterns.
- To visualize key factors affecting used car prices.
- To build an interactive dashboard for better decision-making.

4.METHODOLOGY:

The methodology outlines the step-by-step process followed in the project, from data collection to final analysis and visualization. It ensures a structured approach to achieve the project's objectives by using various tools and techniques.

WEB SCRAPING:

- Data was gathered through web scraping from reputable used car websites.
- Important attributes collected include:
    - Brand
    - Price
    - Fuel Type
    - Manufacturing Year
    - Distance Driven
    - EMI
    - City

DATA CLEANING AND PREPARATION:

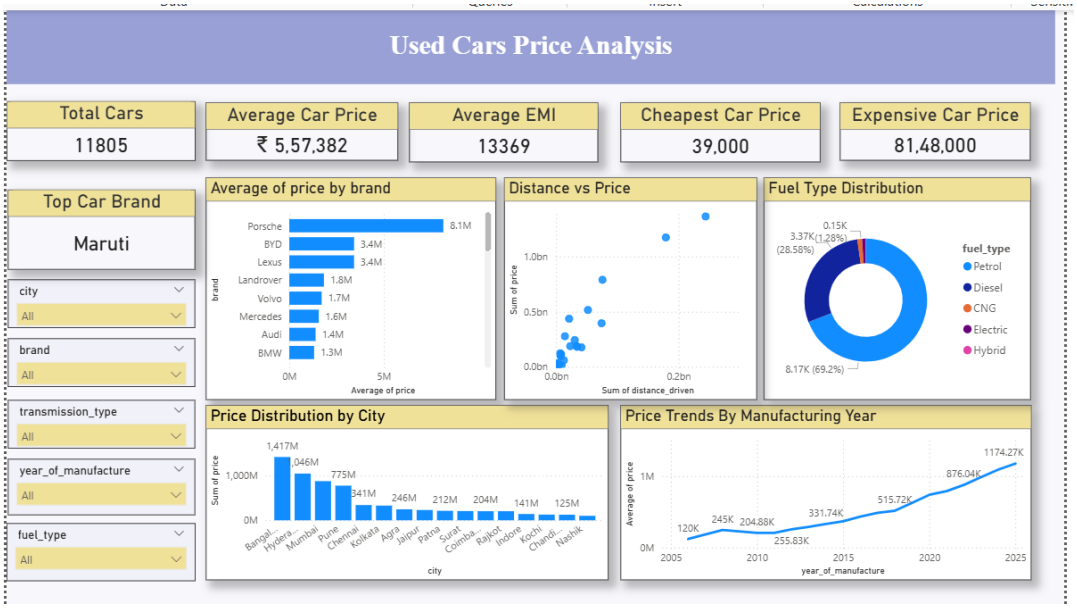
- Removed duplicate records.
- Handled missing and inconsistent values.

- Eliminated unrealistic price outliers.
- Converted columns into appropriate data types.
- Standardized brand names and fuel categories.
- **Tools Used:** Jupyter Notebook

DATA VISUALIZATION

Power BI were used to create dashboards showing:

- Average Price by Brand
- Distance vs Price
- Fuel Type Distribution
- Price Trends by Manufacturing Year
- Price Distribution by City
- **Tools Used: Power BI**



5.KEY FINDINGS

- Used car prices vary significantly across different brands, with premium brands maintaining higher resale value.
- Petrol cars dominate the used car market compared to diesel and other fuel types.

- Cars with lower kilometers driven are generally priced higher than high-usage vehicles.
- Newer manufacturing year cars command higher prices in the resale market.
- EMI availability increases affordability and attracts more buyers in mid-price segments.

6.KEY INSIGHTS

- Buyers can get better value by choosing reliable mid-range brands instead of premium brands.
- Petrol cars are the most practical and easily available option for urban buyers.
- Selecting cars with moderate mileage offers a balance between price and condition.
- EMI options make higher-priced used cars accessible to a wider audience.
- Data-driven analysis helps identify fair pricing and avoid overvalued listings.

7.CHALLENGES FACED

- Extracting data through web scraping was complex due to dynamic website structures.
- Some listings contained missing or inconsistent values.
- Detecting and removing price outliers required careful filtering.
- Combining datasets from multiple cities required additional preprocessing.
- Designing a dashboard that was both informative and visually appealing required multiple iterations.

8. CONCLUSION

The Used Cars Price Analysis project successfully identified the major factors influencing used car prices. By leveraging web scraping, data preprocessing, and visualization techniques, the project converted raw data into actionable insights.

The analysis helps potential buyers understand market trends, compare options effectively, and make cost-efficient purchasing decisions. Additionally, the dashboard provides a user-friendly interface for exploring pricing patterns.

9. REFERENCES

- **Cars24 – https://www.cars24.com**